

## Additional DNS services

### *Host aliasing*

- multiple names map to a single IP address
- one “official” (i.e., *canonical*) name + multiple “nicknames” (i.e., *aliases*)
- used to deploy *virtual hosting*

### *Mail server aliasing*

- a domain name (i.e., alias) maps to one or multiple mail servers (e.g., **@universitadipavia.it**, **@unipv.it**)
- name easy to type/remember by users
- flexibility in the mail service management
- possibility to use the same name for different services

### *Load distribution* across replicated servers (e.g., for streaming services)

- one canonical name maps into multiple IP addresses
- rotation of the addresses provided by DNS
- automatic load balancing

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## DNS

Two main components:

- 1. a *distributed database* implemented on many *geographically distributed servers* characterized by a *hierarchical organization*
2. an application layer *protocol* based on a *client/server model* over *UDP* protocol (on port 53) used to query the database

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## DNS architecture

DNS architecture is built around *Name Servers* that store the distributed databased

Hierarchical organization of the Name Servers in an inverted tree structure (max tree depth=128)

Each node has a label; sibling nodes must have unique labels

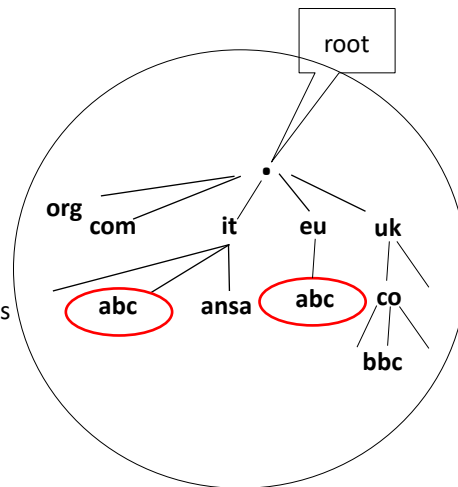
Three main classes of nodes:

- Root
- Top Level Domain (TLD)
- Second Level Domain

A subtree (or a node) denotes a “zone” (i.e., an administrative space)

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*A domain name is a UNIQUE sequence of labels (separated by a dot) from a node to the root*

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## Root Name Servers

The Root is the entry node of the DNS tree identified by a dot “.” (invisible character)

**13** Root Name Servers distributed worldwide and belonging to a reserved domain (i.e., **root-servers.net**) and labelled with a letter from **a** to **m**

The choice of 13 was mainly technical (i.e., answers to DNS queries could fit a single UDP datagram)

Root Name Servers are operated by 12 independent organizations

Root Name Servers are a *very critical piece* of the Internet infrastructure

Many locations/instances of each Root Name Server to ensure resilience and robustness

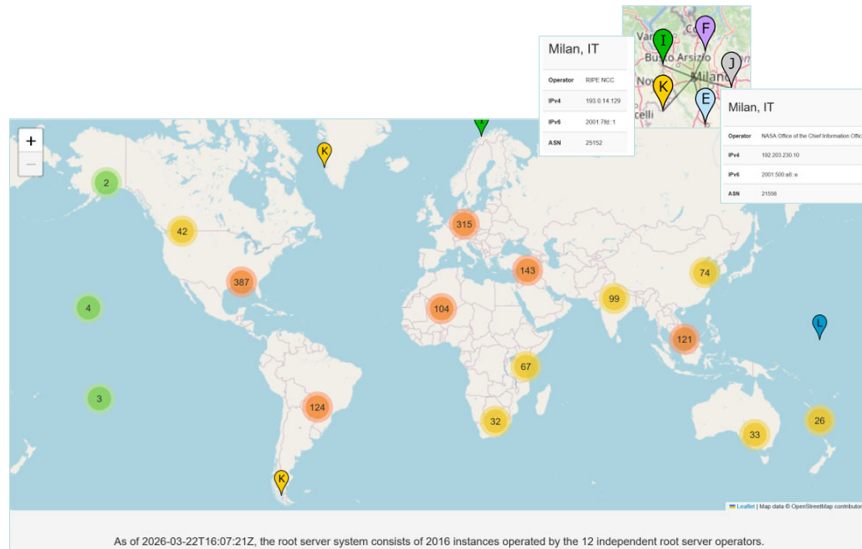
More than 2,000 instances around the world (as of today)

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## Where Root Name Server instances are located



[Source: <https://www.root-servers.org>]

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## List of Root Name Servers

| HOSTNAME           | IP ADDRESSES                      | MANAGER                                 |
|--------------------|-----------------------------------|-----------------------------------------|
| a.root-servers.net | 198.41.0.4, 2001:503:ba3e::2:30   | VeriSign, Inc.                          |
| b.root-servers.net | 199.9.14.201, 2001:500:200::b     | University of Southern California (ISI) |
| c.root-servers.net | 192.33.4.12, 2001:500:2::c        | Cogent Communications                   |
| d.root-servers.net | 199.7.91.13, 2001:500:2d::d       | University of Maryland                  |
| e.root-servers.net | 192.203.230.10, 2001:500:a8::e    | NASA (Ames Research Center)             |
| f.root-servers.net | 192.5.5.241, 2001:500:2f::f       | Internet Systems Consortium, Inc.       |
| g.root-servers.net | 192.112.36.4, 2001:500:12::d0d    | US Department of Defense (NIC)          |
| h.root-servers.net | 198.97.190.53, 2001:500:1::53     | US Army (Research Lab)                  |
| i.root-servers.net | 192.36.148.17, 2001:7fe::53       | Netnod                                  |
| j.root-servers.net | 192.58.128.30, 2001:503:c27::2:30 | VeriSign, Inc.                          |
| k.root-servers.net | 193.0.14.129, 2001:7fd::1         | RIPE NCC                                |
| l.root-servers.net | 199.7.83.42, 2001:500:9f::42      | ICANN                                   |
| m.root-servers.net | 202.12.27.33, 2001:dc3::35        | WIDE Project                            |

[Source: IANA]

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## c.root-servers.net Root Name Server

A B C D E F G H I J K L M

Operator Cogent Communications

Locations Sites: 13

Bratislava, SK Chicago, US Frankfurt, DE Los Angeles, US Madrid, ES New York, US Paris, FR Queretaro, MX Rio de Janeiro, BR Singapore, SG Sydney, AU Tokyo, JP Washington DC, US

IPv4 192.33.4.12

IPv6 2001:500:2::c

ASN 2149

[Source: <https://www.root-servers.org>]

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## 1.root-servers.net Root Name Server

A B C D E F G H I J K L M

Operator ICANN

Locations Sites: 140

Abidjan, CI Alexandria, AU Almaty, KZ Amman, JO Anchorage, US Ancona, IT Arlington County, US Asuncion, PY Atlanta, US Baku, AZ Bangkok, TH Beijing, CN Beirut, LB Bekasi, ID Belem, BR Belo Horizonte, BR Benghazi, LY Biantyre, MW Bogota, CO Brasilia, BR Brazzaville, CG Brisbane, AU Brussels, BE Caceres, BR Cairo, EG Callao, PE Campinas, BR Cape Town, ZA Carire, BR Carolina, PR Changsha City, CN Chicago, US Chisinau, MD Christchurch, NZ CIUDAD DEL ESTE, PY Cochabamba, BO Colombo, LK Curitiba, BR Dammam, SA Dar es Salaam, TZ Dubai, AE El Prat de Llobregat, ES Erbil, IQ Ezeiza, AR Florianopolis, BR Fortaleza, BR Hagatna, GU Haikou, CN Hamburg, DE Harare, ZW Helsinki-Yantai, FI Heraklion, GR Hontare, BR Honolulu, US Incheon, KR Islamabad-Rawalpindi, PK Jaipur, IN Johannesburg, ZA Kalamazoo-Battle Creek, US Kiev, UA Kolonia, FM Koror, PW Kuning City, CN Kuwait City, KW Latina, IT Lawrence, US Leeds-Bradford, GB Londrina, BR Longwood, US Los Angeles, US Madrid, ES Mahé, SC Majuro, MH Mandalay, MM Mangere, NZ Mascot, AU Maseru, LS Melbourne, AU Minsk, BY Monterrey, MX Montevideo, UY Moroni, KM Moscow, RU Mumbai, IN Nadi, FJ Nairobi, KE Natal, BR New Taipei, TW Noumea, NC Odessa, UA Otepén, RO Ouagadougou, BF Papeete, PF Paris, FR Perth, AU Plaisance, MU Port Moresby, PG Porto, PT Porto Alegre, BR Prague, CZ Punta Caucedo, DO Quito, EC Rabat, MA Ramallah, PS Ranchi, IN Rarotonga, CK Reno, US Reston, US Rio de Janeiro, BR Rostov-on-Don, RU Saint Petersburg, RU Salvador, BR Sana'a, YE San Jose, US San Jose, CR San Juan, PR San Miguel de Tucuman, AR San Salvador, SV Santiago, CL Sao Paulo, BR Seeb, OM Semey, KZ Shanghai, CN Singapore, SG Sofia, BG Stockholm, SE Suva, FI Tampere, FI Toronto, CA Tripoli, LY Uberlandia, BR Vancouver, CA Wilmington, US Windhoek, NA Winnipeg, CA Wuhan, CN Xining, CN Yangon, MM Yogyakarta, ID Zhengzhou, CN

IPv4 199.7.83.42

IPv6 2001:500:9f::42

ASN 20144

Legend

IPv6 Enabled Global IPv4 Only Global IPv6 Enabled Local IPv4 Only Local

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## Root Name Servers implementation

Two addresses, i.e., one IPv4 and one IPv6, associated with all instances of a given Root Name Server (e.g., 192.36.98.17 and 2001:7fd::1 for Root NS i)

Anycasting scheme is a routing mechanism used to make instances collectively behave as a single server (e.g., routers choose the “best” instance according to their routing tables) [RFC 1546 “Host Anycasting Service” published in 1993]

Improvements in terms of

- reliability
- scalability
- security

Not every Root Name Server is “anycasted”, i.e., globally replicated

Physical security of Root Name Servers is as important as cyber security

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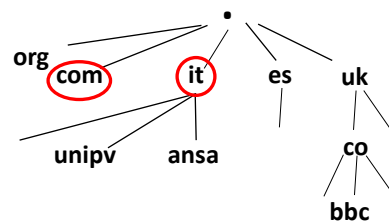
## Top Level Domains

Top Level Domain nodes are the children of the Root

These nodes are identified by the right most label of a hostname or domain name

Examples:

poldo.unipv.it, google.com



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## Types of TLDs

In 1998 the Internet Corporation of Assigned Names and Numbers (ICANN) was established to coordinate the Internet naming system

Two types of TLD nodes:

1. **generic (gTLD)** identified by a string of at least three characters (e.g., **info**, **com**, **eco**, **google**, **net**)
2. **country-specific (ccTLD)** identified by a string of two characters according to the ISO standard 3166 "*International standard for country codes and codes for their subdivisions*" (e.g., **it**, **es**, **ir**, **tr**)

Since 2009 Internationalized Domain Names (IDNs) enable people around the world to use domain names in local languages and scripts (e.g., Arabic, Persian, Chinese)

Domain names start with **xn-** suffix and are encoded with Unicode standard

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## gTLD

Two categories of gTLD:

- *sponsored* by organizations/stakeholders (e.g., **edu**, **mil**, **gov**, **museum**)
- *unsponsored*, i.e., operated by policies established by the Internet community (ICANN) (e.g., **com**, **net**, **org**)

In 2011 ICANN launched the "*New gTLD program*" opening the possibility to everybody (mainly private/public organizations) to register (for a fee) a gTLD name

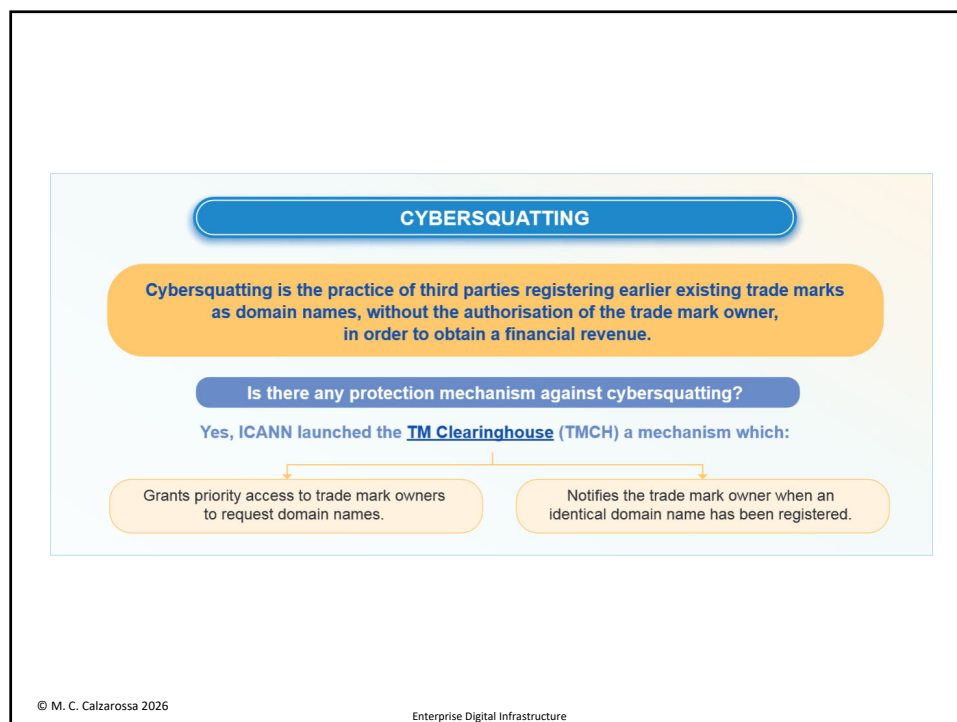
As of today, [1,436 TLDs](#) (gTLD & ccTLD) have been registered

Many companies registered generic TLD names to protect trademarks and avoid *domain name squatting* or *cybersquatting*, that is, an illegal activity of registering gTLDs to make profit from reselling them, selling ads or abusing the domain

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## ccTLD

About 250 ccTLDs operated by organizations (e.g., ISPs, telco companies, universities, research centers) under the “control” of IANA (Internet Assigned Numbers Authority)

ccTLD should be reserved to companies, organizations and individuals based in the country

IANA established the ccTLD **it** in December 1987 and assigned its management to Nation Research Council of Italy – CNR (i.e., **registro.it**)

**it** is currently operated by IIT-CNR at Pisa and can be used by people/ companies based in Italy, EU, Vatican City, San Marino and Switzerland

**eu** was created in 2006 to promote the European dimension and it can be used by companies, institutions and citizens of EU

Some ccTLD are used as “generic” (e.g., **me**, **am**, **fm**, **tv**)

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## Usage of ccTLDs

**Countries that have cashed in on their top-level domain names**

Tuvalu is a tiny island nation of 11,000 people. It's cashing in thanks to Twitch.

**How the tiny Caribbean island of Anguilla has turned the AI boom into a digital gold mine**

Alexander Lee  
November 23, 2019 at 10:56 a.m. EST

***The A.I. Boom Makes Millions for an Unlikely Industry Player: Anguilla***

This small Caribbean territory brought in \$32 million last year, more than 10 percent of its G.D.P., from companies registering domain addresses that end in .ai.

